

Creating an R Project: Step-by-Step

Purpose. When working in RStudio, it is good practice to organize all files associated with a coding project in one dedicated folder. This might include R scripts, data files, figures, reports, and other project materials.

Important: Create the folder first, then create an R Project using that existing folder.

Part 1: Create a Folder for Your Project

First, decide where you want to store the files associated with your project. This might be a folder on your Desktop, Documents, a dedicated courses or research folder, a lab folder, a cloud-synchronized folder, or another location you have designated for storing coding projects. Choose a location that makes sense for you and where you intend to keep the project long-term.

Mac

1. Open Finder.
2. Navigate to the location where you want to store your project.
3. Create a new folder: Right-click (or Control-click) in an empty area and select New Folder.
4. Give the folder a descriptive name, such as:

Marine_Data_Analysis

Windows/PC

5. Open File Explorer.
6. Navigate to the location where you want to store your project.
7. Create a new folder: Right-click in an empty area and select New → Folder.
8. Give the folder a descriptive name, such as:

Marine_Data_Analysis

Your folder might now look something like:

Your designated project location
└─ *Marine_Data_Analysis*

Part 2: Open RStudio

Open RStudio on your computer. You should see several panels, including:

- Source — where you write and edit R code
- Console — where R executes commands
- Environment/History — where you can view objects and command history

- Files/Plots/Packages/Help — where you can view files, plots, packages, and help documentation

Part 3: Create an R Project

There are two ways to create an R Project in RStudio. Both methods accomplish the same thing.

Option 1: Use the Project Button

This is often the easiest method.

9. Look in the upper-right corner of RStudio.
10. You should see a button showing either “Project: (None)” or the name of your current project.
11. Click this button.
12. Select New Project...
13. A window will appear asking what type of project you want to create.
14. Select Existing Directory.
15. Click Browse...
16. Navigate to the folder you created in Part 1.
17. Select your project folder.
18. Click Create Project.

RStudio will open the folder as an R Project.

Option 2: Use the File Menu

19. In RStudio, select File → New Project...
20. Select Existing Directory.
21. Click Browse...
22. Navigate to the folder you created in Part 1.
23. Select the folder.
24. Click Create Project.

RStudio will open the folder as an R Project.

Part 4: Confirm That Your Project Is Open

Look at the upper-right corner of RStudio. You should now see the name of your project instead of “Project: (None).”

For example:

Project: Marine_Data_Analysis

You should also see your project folder in the Files panel.

Part 5: Check Your Working Directory

You can confirm that RStudio is using your project folder as its working directory. In the Console, run:

```
getwd()
```

For example, on a Mac:

```
"/Users/YourName/Projects/Marine_Data_Analysis"
```

On Windows:

```
"C:/Users/YourName/Projects/Marine_Data_Analysis"
```

Your path will look different. That is okay! The important thing is that the path points to the folder you created for your project.

Part 6: Create an R Script

Now that your R Project is set up, create an R script where you will write your code.

25. Select File → New File → R Script.
26. A new script will open in the Source panel.
27. Save the script using File → Save As...
28. Save it inside your project folder.

Examples of descriptive script names include:

- data_cleaning.R
- fish_abundance_analysis.R

Part 7: Add Data and Other Project Files

Your project folder should contain all of the files that are needed for your project.

For example:

```
Marine_Data_Analysis  
├─ Marine_Data_Analysis.R  
└─ fish_data.csv
```

As your project becomes more complex, you might organize it into subfolders:

```
Marine_Data_Analysis  
├─ Marine_Data_Analysis.R
```

```
└─ data
  └─ fish_data.csv
└─ figures
  └─ abundance_plot.png
  └─ size_distribution.png
└─ results
  └─ analysis_results.csv
```

Why Use an R Project?

- 1. It keeps your files organized.** All of the files associated with a project can be kept together.
- 2. It makes file paths easier.** For example, if your data file is in your project folder, you can write `read.csv("fish_data.csv")` instead of specifying the entire path.
- 3. It makes your code more reproducible.** Someone else can open your project and, assuming they have the necessary files and packages, run your code without having to change all of your file paths.
- 4. It allows you to work on multiple projects.** You can have separate R Projects for different research projects, analyses, courses, or other coding activities.

For example:

```
Projects
└─ Marine_Data_Analysis
└─ Coral_Health_Project
└─ Parrotfish_Genetics
└─ Teaching_Data_Visualization
```

Common Mistakes to Avoid

Don't create a new folder when you already created one

If you have already created a folder, choose Existing Directory when creating the R Project. Do not choose New Directory.

Otherwise, you may accidentally create something like:

```
Projects
└─ Marine_Data_Analysis
  └─ Marine_Data_Analysis
    └─ analysis.R
```

Don't save your script somewhere else

Once you have created your R Project, save your R scripts inside the project folder. Avoid keeping the script in one location and the data in another.

Instead, keep the project files together:

Projects

```
└─ Marine_Data_Analysis
    └─ analysis.R
    └─ fish_data.csv
```

Don't use `setwd()` to manually set your working directory

You may see code like this online:

```
setwd("C:/Users/YourName/Projects/Marine_Data_Analysis")
```

When you are working within an RStudio Project, you generally should not need to use `setwd()`. The R Project automatically sets the working directory to the project folder.

Quick Checklist

- ☐ I created a dedicated folder for my project.
- ☐ I chose a location where I want to store the project long-term.
- ☐ I created an R Project using Existing Directory.
- ☐ My R Project is connected to the folder I created.
- ☐ The project name appears in the upper-right corner of RStudio.
- ☐ I created and saved my R script inside the project folder.
- ☐ My data and other project files are stored within the project folder.
- ☐ `getwd()` shows my project folder.
- ☐ I understand that I generally do not need to use `setwd()` when working in an R Project.

Once you've completed these steps, you're ready to start coding!